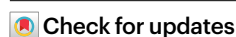


Regulating phase homogeneity by self-assembled molecules for enhanced efficiency and stability of inverted perovskite solar cells

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Heterogeneity in transporting interfaces and perovskites poses a substantial challenge in improving the efficiency of perovskite solar cells from small to large scales, a key barrier to their commercial use. Here we find that the amorphous phases of self-assembling molecules (SAMs) can realize a more homogeneous perovskite growth. Hyperspectral analysis confirms a narrower and blueshifted photoluminescence peak distribution in perovskite/amorphous SAMs. Additionally, fluence-dependent time-resolved photoluminescence reveals a reduced trap-assisted recombination rate of $0.5 \times 10^6 \text{ s}^{-1}$ in amorphous-SAM-based perovskite films. This improvement translates to p–i–n structured perovskite solar cells achieving an efficiency of 25.20% (certified at 24.35%) over a one-square-centimetre area. These cells maintain nearly 100% efficiency after 600 h of 1-sun maximum power point tracking under the ISOS-L-1 protocol, and retain 90% of their initial efficiency after 1,000 h, as evaluated by the ISOS-T-2 protocol.

Recent trends in perovskite solar cell (PSC) research have shown a growing preference for the inverted (p–i–n) architecture, while progressively narrowing the gap in power conversion efficiency (PCE) compared to the normal structured (n–i–p) structure^{1–4}. An important factor in this efficiency improvement is the use of self-assembled molecules (SAMs) as the hole-transporting material (HTM)^{4–10}. These HTM SAMs generally comprise a hole-transporting component, anchoring groups and spacers, with the anchoring group (for example, phosphoric acid)

chemically bonding to metal oxides or transparent conducting oxide (TCO) substrates¹¹.

In perovskite photovoltaics, SAM deposition methods commonly make use of rapid solution processing, diverging from the traditional ‘self-assembled monolayer’ concept and complicating the achievement of a classic monolayer morphology¹². These SAMs tend to aggregate or crystallize, driven by weak Coulomb forces and strong van der Waals interactions, particularly π – π interactions^{13–16}. Unanchored SAMs often

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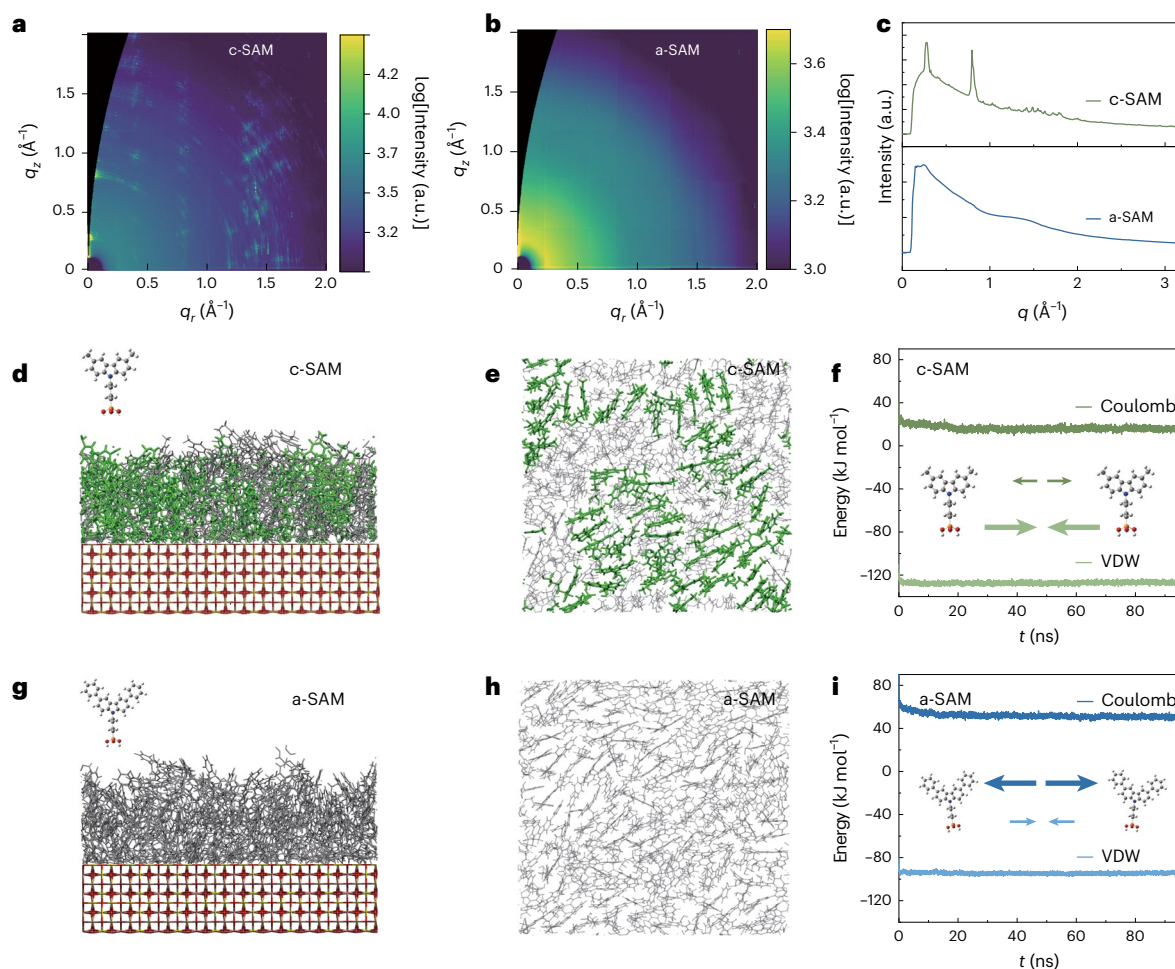


Fig. 1 | Molecule structure and morphology on TCOs. **a**, 2D GIWAXS image of c-SAM deposited on TCO substrates transferred to q -space. **b**, 2D GIWAXS image of a-SAM deposited on TCO substrates transferred to q -space. **c**, Radial GIWAXS profiles integrated over all azimuthal angles (pseudo-X-ray diffraction profiles) for c-SAM and a-SAM. **d**, MD simulation results of a side view of c-SAM absorbed on TCOs (50 ns, 6 nm $\times$ 6 nm). **e**, MD simulation results of a top view of c-SAM

absorbed on TCOs (the crystallized domain is highlighted in green; 50 ns).

f, MD-simulated Coulomb and van der Waals interaction energies of c-SAM.

g, MD simulation results of a side view of a-SAM absorbed on TCOs (50 ns). **h**, MD simulation results of a top view of a-SAM absorbed on TCOs (50 ns, 6 nm $\times$ 6 nm).

i, MD-simulated Coulomb and van der Waals interaction energies of a-SAM.

remain on top of anchored ones, forming nanometre-thick stacks with sparse molecular packing. This scenario parallels the dye-sensitized and organic-solar-cell fields, where suboptimal crystallization and non-uniformity correlate with moderate solar-cell performance^{13,17,18}. However, details regarding the surface packing and morphological growth of SAMs on TCO substrates in perovskite devices, particularly regarding homogeneity over large areas, remain poorly understood.

As endorsed by the photovoltaic community, the cell efficiency for areas greater than 1 cm² is a more relevant metric for module-level potential. This standard is reflected in the inclusion of such cells in table 1 of the solar cell efficiency tables¹⁹. Over the past 14 years, normal structured PSCs have led efficiency advancements^{2,20}. However, inverted structured perovskites, notable for their enhanced stability and scalability, are increasingly a focus in both academic and industrial developments of perovskite single-junction and tandem technologies^{21,22}. Thus, the efficiency gap between normal and inverted structured PSCs, particularly for large, commercially relevant active areas, is a critical challenge in the field^{3,23,24}.

In this study, evidenced by a comparative analysis of (4-(3,6-dimethyl-9H-carbazol-9-yl)butyl)phosphonic acid⁵ (Me-4PACz, a crystallized SAM (c-SAM)) and (4-(3,6-diphenyl-9H-carbazol-9-yl)butyl)phosphonic acid (Ph-4PACz, an amorphous SAM (a-SAM)), we show that steric hindrance and intermolecular interactions are the means for realizing

uniform amorphous phases in SAMs, thereby facilitating homogeneous perovskite growth. Grazing-incidence wide-angle X-ray scattering (GIWAXS) results reveal a fully amorphous phase, and molecular dynamics (MD) simulations confirm the homogeneity distribution of Ph-4PACz, leading to a more focused and blueshifted photoluminescence (PL) peak distribution in perovskite/a-SAM films. Moreover, fluence-dependent time-resolved PL (TRPL) analysis indicates a lower trap-assisted recombination rate of 0.5×10^6 s⁻¹ in a-SAM-based perovskite films^{25,26}. This advance of high homogeneity and low trap density results in p-i-n structured PSCs achieving a groundbreaking efficiency of 25.20% (with a certified maximum power point tracking (MPPT) efficiency of 24.35%) across a one-square-centimetre area. This 1-cm² efficiency is a great advance for inverted structured perovskite cells, surpassing normal structured cells and being recognized in the solar cell efficiency tables. Furthermore, a-SAM-based PSCs demonstrate stability that is superior to that of c-SAM, maintaining nearly 100% of their initial PCE after 600 h of MPPT under the ISOS-L-1 protocol and 90% under 85 °C thermal stress following the ISOS-T-2 protocol.

Results and discussion

Uniformity of SAM layers

To examine the structural properties of the SAM HTMs, GIWAXS was initially used to assess the crystallinity of the state-of-the-art

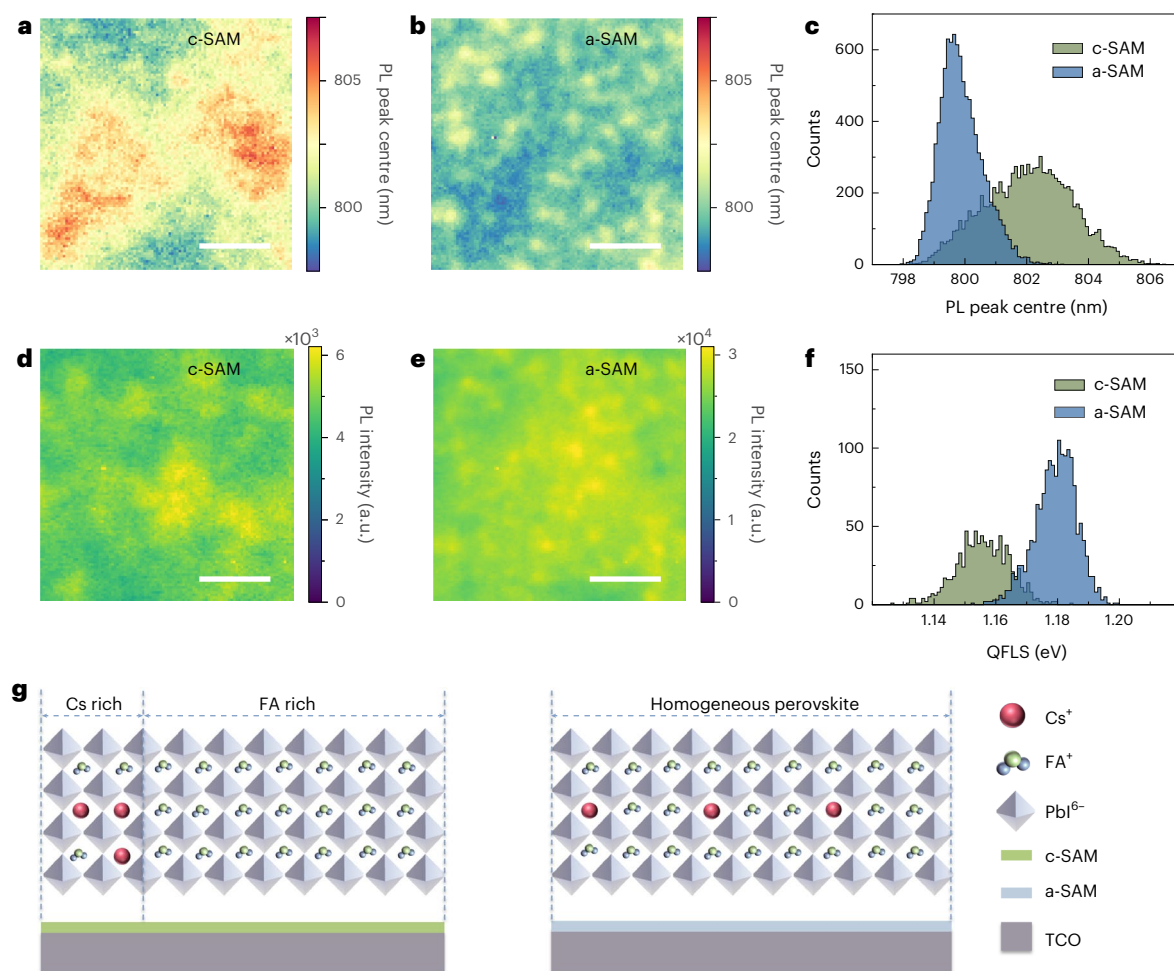


Fig. 2 | Uniformity of perovskite film on c-SAM and a-SAM. a, b, PL peak distribution extracted from the measured area for the encapsulated perovskite film on c-SAM (a) and a-SAM (b). **c,** Histogram of PL peak positions of the perovskite on c-SAM and a-SAM emission. **d, e,** PL intensity maps of the encapsulated perovskite film on c-SAM (d) and a-SAM (e). The images were

acquired in ambient conditions with a $\times 20$ objective, 405-nm incident light (λ_{exc}) and an illumination flux of 100 mW cm^{-2} . Scale bars, $10 \mu\text{m}$. **f,** Histogram of QELS for the perovskite film on c-SAM and on a-SAM. The values were extracted from fits to local absolute PL spectra. **g,** A schematic plot showing that a-SAM can realize more homogeneous growth for the pure perovskite phase.

Me-4PACz. The GIWAXS results show that Me-4PACz exhibits discrete Bragg spots, indicating pronounced crystallinity, as shown in Fig. 1a,c, with noticeable orientation. To further delve into the impact of crystallinity on morphology, MD simulations were conducted to analyse the molecular distribution on TCO substrates. Figure 1d,e presents a snapshot from a 50-ns MD simulation. These images depict the formation of more ordered domains, highlighted in green, by the molecule Me-4PACz. These structured domains show a more organized molecular arrangement. The muted colour amorphous domains appear more chaotic and less organized, in contrast with the structured regions. The crystallization of SAMs is dominantly determined by intermolecular forces and molecule geometry. These crystallized domains arise from the interplay of van der Waals forces (dispersion, dipole–dipole and dipole-induced dipole interactions) and Coulomb (electrostatic) interactions among the carbazole moieties^{27,28}. The quantification of these interactions through MD simulations reveals that the overall Coulomb interaction is a repulsive force of -10 kJ mol^{-1} , in contrasted with an overall attractive van der Waals interaction of 130 kJ mol^{-1} . This suggests that strong van der Waals forces predominantly govern the interactions in Me-4PACz. Furthermore, the results from MD simulations align with the two-dimensional (2D) GIWAXS patterns of Me-4PACz, which demonstrate distinct orientation and crystallinity.

Steric hindrance and intermolecular interactions among molecules are known to influence the crystallinity of SAMs^{29,30}. The SAM Ph-4PACz was synthesized by attaching an extended benzene ring to a carbazole core through a single bond, effectively replacing the methyl group in c-SAM while retaining the same spacer³¹. This design, featuring a rotatory phenyl group, aims to augment steric hindrance and diminish attractive intermolecular interactions between molecules. The comprehensive synthesis route for Ph-4PACz, along with ¹H NMR and ¹³C NMR, as well as matrix-assisted laser desorption/ionization-time of flight (MALDI-TOF) mass spectra of an intermediate product are detailed in Supplementary Figs. 1–10. MD simulations (Fig. 1i) indicate an increase in Coulomb repulsive force to 50 kJ mol^{-1} and a decrease in van der Waals force to 90 kJ mol^{-1} for Ph-4PACz, combining the large steric hindrance introduced by the extended benzene ring linked with a rotatory single bond, meaning that there is no crystallized phase observed in the MD-simulated SAM morphology (Fig. 1g,h). These changes foster a fully amorphous phase in the Ph-4PACz layer on TCO substrates (Fig. 1b,c), thereby enhancing the uniformity of the hole-transport layer. In the following, we use c-SAM to refer to Me-4PACz and a-SAM for Ph-4PACz.

Both c-SAM and a-SAM exhibit an increased intensity in the out-of-plane direction, a feature potentially beneficial for charge-carrier transport in PSCs (Supplementary Fig. 11). Despite having

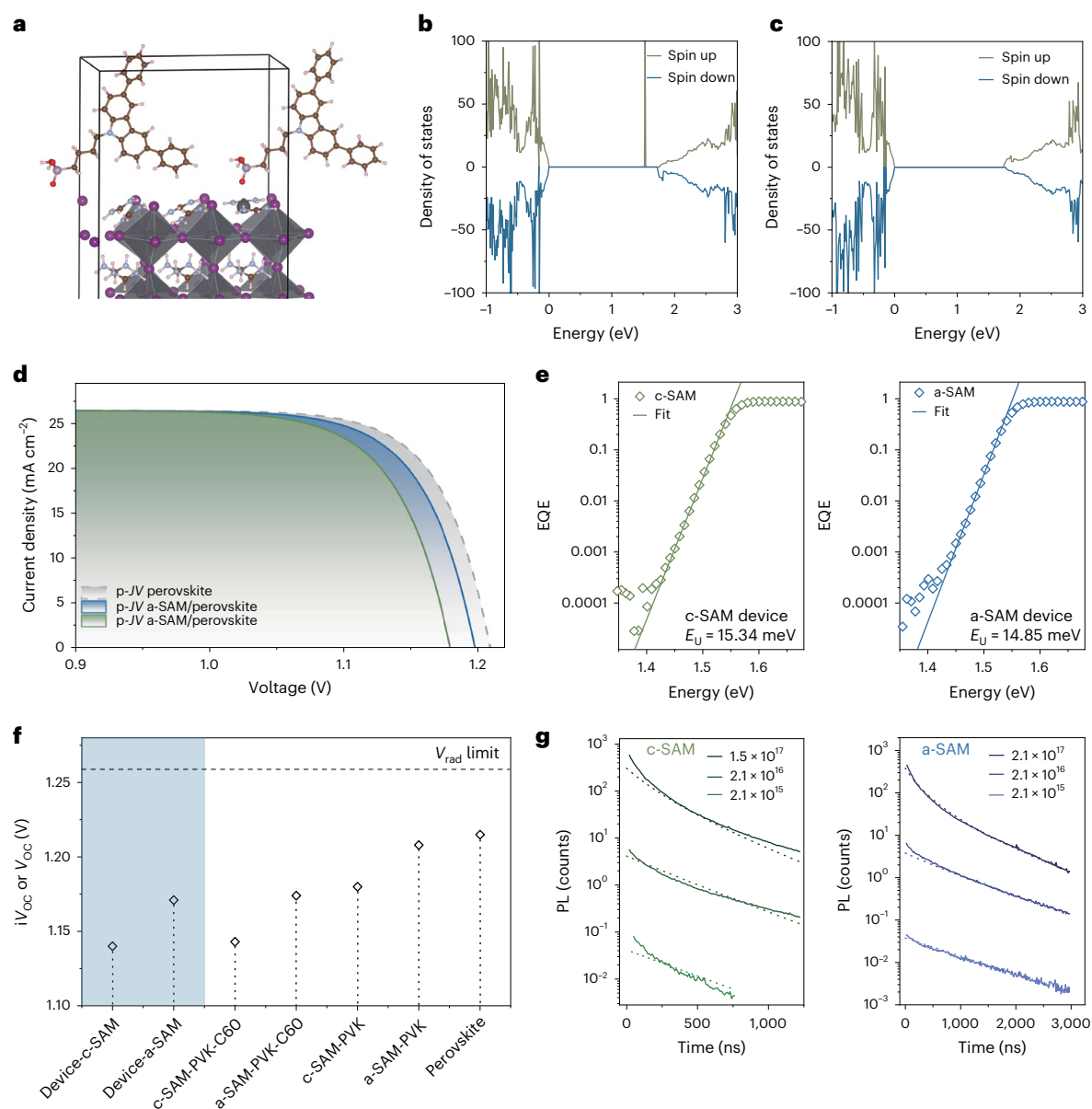


Fig. 3 | Energy loss and charge-carrier dynamic analysis. **a**, DFT optimized structure of the passivated FAPbI₃ surface with a-SAM. **b**, Calculated density of states of FAPbI₃ with Pb_i defects. **c**, Calculated density of states of FAPbI₃ with Pb_i defects passivated by a-SAM. **d**, Pseudo-JV (p-JV) curves obtained from intensity-dependent V_{OC} and QFLS measurements on the neat perovskite and the SAMs/perovskite. **e**, High-resolution EQE spectra of c-SAM and a-SAM PSCs. E_U is Urbach

energy. **f**, The iV_{OC} of the individual films. **g**, Excitation-dependent PL kinetics of perovskite on c-SAM and a-SAM normalized by intensity at time zero. The dashed lines are globally modeled PL decay curves for subtracting k_1 and k_2 . Initial carrier densities are calculated as outlined in Supplementary Information. Inset values: initial PL intensity versus injected carrier density at time zero, n_0 .

a similar roughness and morphology on TCOs (Supplementary Fig. 12), kelvin probe force microscopy (KPFM) further corroborated the enhanced uniformity of these SAMs on TCOs from an electrical standpoint (Supplementary Fig. 13). The potential root-mean-square (R_q) of a-SAM (0.073 V) is markedly lower than that of c-SAM (0.219 V). This indicates a substantial improvement in the surface potential uniformity of the SAM layer when a-SAM is used.

Characterizations of perovskite homogeneity

The growth of perovskite on SAMs with varying degrees of inhomogeneity results in distinct spatial and spectral variations in the PL peak maximum of the perovskite⁸. To investigate the impact of different SAM phases on the heterogeneity of perovskite materials, a hyperspectral PL mapping was performed for the same perovskite material deposited on

c-SAM and a-SAM (Fig. 2a,b)³². The perovskite layer on a-SAM exhibits a narrow distribution in PL peak positions (Fig. 2b), in contrast to the broader distribution observed with c-SAM (Fig. 2a). This broader range in c-SAM is accompanied by redshifted PL (Fig. 2c), potentially indicative of segregated or defective phases arising from the diverse surface energy distributions within the crystallized SAM³³. Conversely, the peak positions of perovskite on a-SAM were predominantly centred around 800 nm, closely aligning with the PL emission of the reference material formamidinium-cesium perovskite Cs_{0.1}FA_{0.9}PbI₃ (ref. 34).

Phase heterogeneity also influences the optoelectronic properties of perovskite materials on different SAMs^{35,36}. The perovskite film on a-SAM exhibited a PL intensity ten times higher and with a narrower distribution than for the perovskite film on c-SAM (Fig. 2d,e), while X-ray diffraction (XRD) and scanning electron microscopy (SEM)

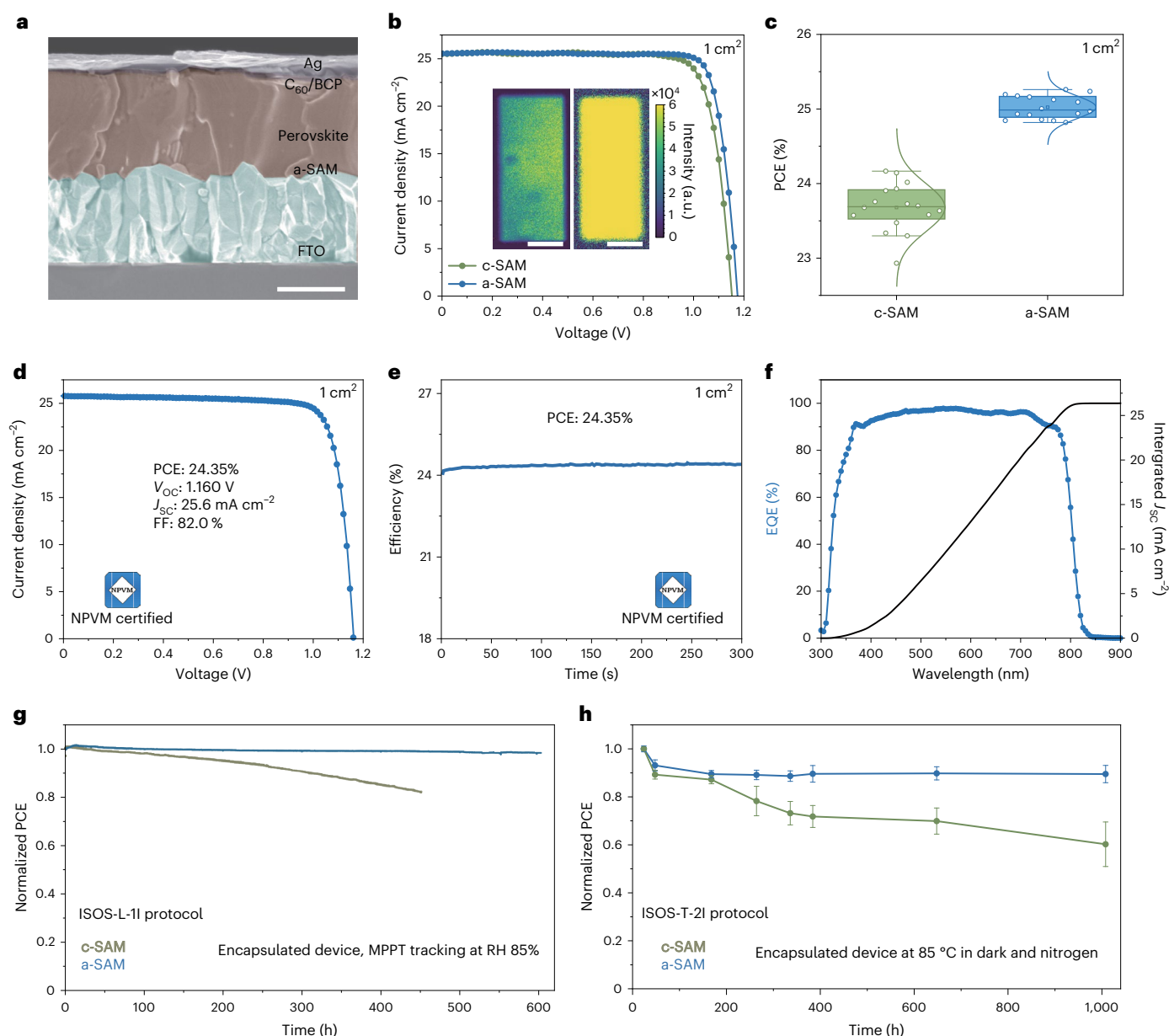


Fig. 4 | Device characterizations. **a**, Cross-sectional SEM of the whole PSC with the architecture of FTO/a-SAM/Cs_{0.1}FA_{0.9}PbI₃/C₆₀/BCP/Ag. Scale bar, 500 nm. **b**, *J*-*V* scans of 1-cm² c-SAM and a-SAM PSCs. Inset: electroluminescence mapping of c-SAM (left) and a-SAM (right) PSCs. Scale bars, 5 mm. **c**, PCE statistical distribution of 1-cm² PSCs with c-SAM and a-SAM (16 devices for each group). In the boxplots, the mean (open square), median (centreline) and 25th and 75th percentiles (box bounds) are shown. **d**, *J*-*V* scan measurement of the 1-cm²

device recorded by NPVM. **e**, Stabilized output measurement of the 1-cm² device recorded by NPVM. **f**, Certified EQE spectra of 1-cm² a-SAM PSCs. The integrated *J*_{sc} is 26.3 mA cm⁻². **g**, Continuous MPPT of PCE for encapsulated PSCs with c-SAM and a-SAM under the protocol of ISOS-L-II. RH, relative humidity. **h**, Thermal stability measurement of PCE for PSCs with c-SAM and a-SAM under the protocol of ISOS-T-2I (six devices for each group, data are presented as mean values ± s.d.).

results show a similar structure and morphology for the perovskite film (Supplementary Figs. 14 and 15). We also extracted the quasi-Fermi level splitting (QFLS) of the perovskite on different SAMs, which is directly related to the open-circuit voltage (*V*_{OC}) of a solar cell (Fig. 2f and Supplementary Fig. 16). The perovskite on a-SAM showed an average QFLS of 1.18 V, which is higher than for c-SAM (1.15 V). This suggests a more homogeneous and lower-defect growth of the perovskite layer on a-SAM, which could be advantageous for the performance of perovskite in a large area (Fig. 2g).

*V*_{OC} loss analysis of PSCs

First-principles electronic structure calculations were performed to assess the passivation effects of the a-SAM on common deep-level

defects on the perovskite grain surface, such as interstitial lead (Pb)^{23,37}. It was found that a-SAM exhibits a different preferred absorption geometry compared to 2PACz, as reported previously¹². In addition to the bond between the phosphoryl group and perovskite, the a-SAM molecule also interacts with the perovskite surface via its extended benzene ring in an optimized density functional theory (DFT) adsorption model (Fig. 3a). This interaction contrasts with the defective states within the bandgap depicted in Fig. 3b, as Fig. 3c reveals an absence of such states, highlighting the lower defects density of perovskite film grown on a-SAM.

To explore the impact of these SAMs on device performance, particularly on parameters such as *V*_{OC}, intensity-dependent steady-state PL measurements were conducted. Pseudo-*J*/*V* curves

were constructed for both c-SAM and a-SAM stacks (Fig. 3d), and the resulting implied open-circuit voltage (iV_{oc}) and pseudo-fill factor (pseudo-FF) values are summarized in Supplementary Table 1. Although perovskite films exhibited the highest simulated photovoltaic performance when deposited on glass substrates, introducing transport layers reduced the inferred device performance due to interface recombination losses, as previously reported³⁸. In perovskite/HTM stacks, the a-SAM-based stack showed a superior pseudo-PCE, reaching 27.61%, with an iV_{oc} of 1.20 V and a pseudo-FF of 86.9%. These metrics surpass those of the c-SAM-based stack, which demonstrated a pseudo-PCE of 27.04%, an iV_{oc} of 1.18 V and a pseudo-FF of 86.5%.

To further assess the voltage losses, we utilized high-resolution external quantum efficiency (Hr-EQE) measurements and then determined the radiative voltage limits to provide a more precise quantification of the voltage losses arising from non-radiative recombination (Fig. 3e)³⁹. The Urbach energy of the a-SAM-based device was determined to be 14.85 meV, which is smaller than the value of 15.34 meV observed for the c-SAM-based device. This result suggests that the a-SAM-based device exhibits fewer defects than the c-SAM-based device. The slight variation in Urbach energy suggests a minimal impact on V_{oc} in the radiative limit, with the V_{rad} of both devices determined to be 1.26 eV (ref. 40). In Fig. 3f, we summarize the V_{oc} , QFLS and radiative-limit voltage values for both c-SAM and a-SAM devices/stacks. The iV_{oc} values derived for a-SAM stacks are consistently higher than for the c-SAM stacks, which agrees with the trend of measured V_{oc} in devices. Considering the comparable radiative limit of a-SAM- and c-SAM-based devices, the primary difference in V_{oc} stems from non-radiative recombination (Supplementary Table 2). Moreover, the FF decrease caused by trap-induced non-radiative recombination is also mitigated, which leads to a 1% absolute FF gain in the c-SAM-based device, as shown in Supplementary Fig. 17.

To support the V_{oc} and FF loss caused by the trap-assisted recombination process, a quantitative analysis of the recombination process using fluence-dependent TRPL spectra was performed to evaluate the defect concentration of perovskite films grown on SAMs with different phases²⁶. These measurements were conducted at different fluences to ensure that the recombination process covered the trap-assisted (k_t) and radiative recombination processes (k_r). Different excitation densities (from $2.1 \times 10^{15} \text{ cm}^{-3}$ to $2.1 \times 10^{17} \text{ cm}^{-3}$) allowed us to fit the k_t accurately (Supplementary Fig. 18). Pure perovskite film and perovskite film on c-SAM have similar trap-assisted recombination rates ($1.3 \times 10^6 \text{ s}^{-1}$; Supplementary Fig. 19 and Supplementary Table 3), but this trap-assisted recombination rate decreases to $0.5 \times 10^6 \text{ s}^{-1}$ for a-SAM-based perovskite films. According to our previous method for deducing the theoretical maximum V_{oc} , the upper limit of V_{oc} can increase from 1.17 to 1.20 V, indicating a less defective perovskite film on a-SAM⁴¹.

Photovoltaic performance of 1-cm² PSCs

Both high-performance c-SAM-PSCs and a-SAM-PSCs are based on a p-i-n architecture consisting of glass/fluorine-doped tin oxide (FTO)/c-SAM or a-SAM/Cs_{0.1}FA_{0.9}PbI₃/C₆₀/bathocuproine (BCP)/Ag as shown by the cross-sectional SEM in Fig. 4a. Figure 4b shows the $J-V$ curve of 1-cm² c-SAM and a-SAM-based devices. With the improved uniformity of a-SAM, the 1-cm² a-SAM-based devices show a maximum PCE of 25.20%, with a V_{oc} of 1.175 V, FF of 84.0% and J_{sc} of 25.6 mA cm⁻², which is the first reported 1-cm² device with a PCE over 25%. In contrast, the c-SAM-based devices show moderate V_{oc} and FF values of 1.153 V and 81.6%, resulting in a PCE of 24.0%. This result is also in line with the PCE statistics shown in Fig. 4c and Supplementary Fig. 20. Notably, there is a discernible disparity in FF between the 1-cm² c-SAM and a-SAM devices (FF_{avg} for the c-SAM device is 81% and for the a-SAM device is 83.5%). We assume that this is because of the uniformity difference between these two SAMs. We sent one 1-cm² PSC to the National

Photovoltaic Industry Metrology Test Center (NPVM) for certification (Supplementary Fig. 21). We achieved a MPPT efficiency of 24.35% (Fig. 4d,e), which is a high certified efficiency for a 1-cm² PSC compared with n-i-p and p-i-n configurations reported in the published literature (Supplementary Fig. 22 and Supplementary Table 4). The certified EQE of the a-SAM-based device is shown in Fig. 4f. The integrated J_{sc} is 26.3 mA cm⁻².

Electroluminescence mapping was also conducted on devices with the same architecture and an active area of 1.70 cm². The mapping, as illustrated in the inset picture of Fig. 4b, reveals several dark regions in the c-SAM device, suggesting the presence of inhomogeneous perovskite components. In stark contrast, the device employing a-SAM demonstrates a substantially higher and more uniform electroluminescence emission intensity across the entire cell area. These findings align with the PL mapping results, pointing towards a substantial reduction in non-radiative recombination at the HTM/perovskite interface. This reduction contributes to the observed enhancements in V_{oc} , FF and overall device stability due to the mitigation of interfacial defects.

Rigorous stability assessments were also conducted, adhering to the ISOS-L-1I and ISOS-T-2I protocols⁴². The encapsulated a-SAM device remarkably retained nearly 100% of its initial efficiency after 600 h of MPPT under continuous one-sun light-emitting diode (LED) illumination at a relative humidity of 85%. Conversely, the initial PCE of the c-SAM device diminished to below 85% after just 400 h under similar conditions (Fig. 4g). Furthermore, the a-SAM-based device impressively preserved 90% of its PCE following 1,000 h of exposure to 85 °C thermal stress, greatly outperforming the c-SAM-based device, which maintained only 56% of its original PCE (Fig. 4h).

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41566-024-01531-x>.

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Methods

Materials

Cesium iodide (CsI, 99.99%) lead chloride (PbCl_2), dimethylformamide (DMF, anhydrous) and 1-methyl-2-pyrrolidinone (NMP, anhydrous) were purchased from Sigma-Aldrich. Formamidinium iodide (FAI) was purchased from Greatcell Solar Materials, and anhydrous lead iodide (PbI_2) and Me-4PACz were purchased from Tokyo Chemical Industry. Silver was purchased from Alfa Aesar, and C_{60} and BCP were purchased from Lumtec. For the synthesis of Ph-4PACz, commercially available reagents were purchased from Admas and used without further purification. Toluene and tetrahydrofuran were freshly distilled before use. Other solvents were used directly.

Device fabrication

The patterned ITO or FTO glass substrates were wiped with soap water, then washed ultrasonically with deionized water, acetone and isopropanol for 30 min each. The substrates were transferred to a nitrogen glovebox after ultraviolet ozone treatment for 15 min or 450-W air plasma treatment for 5 s. The c-SAM (1 mg ml^{-1} in isopropyl alcohol) or a-SAM (1 mg ml^{-1} in isopropyl alcohol) was spin-coated on the substrates at 4,000 r.p.m. for 30 s and heated at 100°C for 10 min. The 1.8 M perovskite ($\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$) stock solution contained 52 mg of CsI, 309.6 mg of FAI, 922 mg of PbI_2 and 55.6 mg of PbCl_2 , which were dissolved in 1.11 ml of mixed solvent (DMF:NMP, vol:vol = 960:150). The 1.8 M perovskite precursor solution was shaken overnight to fully dissolve and then used to prepare perovskite films. In the spin-coating process, the substrate was spun at 5,000 r.p.m. for 40 s with an acceleration of $5,000 \text{ r.p.m. s}^{-1}$, and N_2 gas was blown on top of the spinning substrate after 20 s. The perovskite films were pre-annealed at 70°C for 2 min in the N_2 glovebox and then annealed at 150°C for 10 min in 30% relative humidity. Finally, C_{60} (20 nm)/BCP (8 nm)/Ag (100 nm) layers were deposited to complete the device fabrication. The devices were masked with metal aperture masks (1 cm^2) during the J - V measurements.

Device characterization

J - V measurements of the devices were recorded with a Keithley 2400 source meter under simulated 1-sun AM 1.5G illumination (100 mW cm^{-2}) with an ABET Technologies Sun 2000 solar simulator. The measurements were performed in a N_2 -filled glovebox. The illumination area of the devices was defined by shadow masks of a 1.00-cm^2 area. MPPT measurements of the devices were recorded with an MPP Tracking-4B system (Shenzhen Lancheng Technology) with an LED-simulated AM1.5G spectrum. The measurements were carried out under ambient conditions at 40°C with a relative humidity of ~85%, with the devices encapsulated. EQE measurements were conducted using a Bentham PVE300-IVT system. The LED laser intensity was calibrated with built-in silicon and germanium diodes before measurements. The light was chopped at 137 Hz for the high-resolution EQE measurements and coupled into a Bentham monochromator. The resulting monochromatic light was focused onto the PSC, and its current under short-circuit conditions was fed to a current preamplifier (Stanford SR 570) before it was analysed with a lock-in amplifier (Stanford SR830 DSP). The time constant of the lock-in amplifier was chosen to be 1 s, and the preamplifier's amplification was increased to resolve low photocurrents. The EQE was determined by dividing the photocurrent of the cell by the flux of incoming photons, which was measured using a calibrated Si photodiode. All EQE measurements were performed under ambient conditions at 25°C , with a relative humidity of ~60%, and without encapsulation. Suns- V_{oc} PLQY was conducted with an LP20-32 radiative efficiency meter (QYB). For suns- V_{oc} measurements, the laser was switched on 15 min before the measurement to allow it to warm up and stabilize. The laser was then set to the desired intensities, and the PL measurements were performed. Plotting of the suns- V_{oc} and pseudo- J - V follows previous reports³⁸. For the pseudo- J - V , 85% of

Shockley–Queisser limit current density was used to avoid potential influences for the iV_{oc} comparison.

Fluence-dependent TRPL spectra

TRPL measurements were carried out using a 407-nm pulsed supercontinuum laser as an excitation source. An absorptive 420-nm long-pass filter was used to filter out scattered laser light. The focused PL was detected by a silicon-based single-photon avalanche photodiode (MPD-PDM-PDF). The instrument response time was ~200 ps. For the investigation of recombination behaviour, TRPL spectra were recorded using a gated intensified charge-coupled device (CCD) camera (Andor iStar DH740 CCI-010) connected to a calibrated grating spectrometer (Andor SR303i). The 800-nm emissions from a Ti:sapphire optical amplifier (1-kHz repetition rate, 90-fs pulse width) were frequency-doubled to generate narrow-bandwidth excitation centred at a wavelength of 400 nm. The incident pulse energy was varied from 0.0152 to $1.52 \mu\text{J cm}^{-2}$. Initial excited carrier densities were calculated according to the method of Richter and colleagues²⁵. The effective area of the excitation spot was 1.48 mm^2 .

Hyperspectral microscope characterization

Wide-field hyperspectral microscopy measurements were conducted using the Photon etc. IMA model. We employed $\times 20$ air objective lenses from Olympus (MPLFN), which were specifically chromatic aberration-corrected. We utilized a continuous-wave laser with a wavelength of 405 nm. A dichroic mirror was utilized to filter the excitation laser. The emitted light from the sample was directed onto a volume Bragg grating, which spectrally separated the light and directed it onto a CCD camera. The CCD camera, maintained at 0°C with a thermoelectric cooler, had a resolution of $1,024 \times 1,024$ pixels and operated within a wavelength range of 700–900 nm. By systematically varying the angle of the grating concerning the incident light, we obtained the spectral information for the light emanating from each point on the sample.

A two-step process was implemented for each objective lens to establish the system's calibration and extract the absolute number of photons at each point. First, a calibrated white-light lamp (Ocean Optics) was integrated into an integrating sphere. Simultaneously, the objective lens was also connected to the integrating sphere. By comparing the recorded spectrum of the lamp at each point to the calibrated spectrum, we determined the system's relative sensitivity both spectrally and spatially. Second, a 657-nm laser was directly coupled to the microscope using an optical fibre. The laser's power was meticulously measured at the fibre output with a power meter before coupling to the objective lens. This power measurement of the laser within the system allowed for a direct correlation between the number of counts and the photons at the 657-nm wavelength. By merging this absolute calibration, obtained from the laser measurements, with the relative calibration data derived from the calibrated white-light lamp and integrating sphere, we achieved a comprehensive absolute calibration across the entire spectrum at each sample point.

Suns intensity calculation and QFLS extraction

We followed a series of steps to determine the equivalent number of suns for a specific monochromatic excitation with a given power. First, we utilized an interpolated AM1.5G spectrum and transformed the data from units of spectral irradiance ($\text{W m}^{-2} \text{ nm}^{-1}$) to photons $\text{m}^{-2} \text{ nm}^{-1} \text{ s}^{-1}$. This conversion involved dividing the spectral data by the photon energy at each wavelength point in the spectrum. Next, we integrated the resulting spectrum over the wavelength range spanning from 400 nm to the material's bandgap energy, ~807.8 nm. This integration yielded the total flux of photons with energy above the bandgap, expressed as photons $\text{m}^{-2} \text{ s}^{-1}$. Subsequently, we compared this photon flux to the photon flux originating from the monochromatic excitation incident on the sample, allowing us to calculate the equivalent number of suns. QFLS was extracted based on the method of ref. 38.

GIWAXS methods

We prepared samples by drop-casting 1 mg ml^{-1} Me-4PACz and Ph-4PACz solution in ethanol on 500- μm float-zone silicon wafers, then carried out an automatic drying process. GIWAXS was performed at the PETRA III synchrotron P03 beamline⁴³. A beam ($23 \times 32 \mu\text{m}^2$) with a monochromatic X-ray energy of 11.9 keV (corresponding to 1.044 Å) with a high brilliance impinged the samples at an incidence angle of 0.2° , which is the critical angle of the involved materials, to probe the sample morphology. For GIWAXS, a Lambda 9M detector was used with a sample-to-detector distance of 204.8 mm. Data reduction was performed with INSIGHT software, including typical geometrical and intensity corrections (solid-angle correction, detector-pixel-sensitivity correction, polarization correction and air-attenuation correction)⁴⁴. A Si attenuation of 2.33 mm^{-1} and horizontal polarization of 0.98 were used, and an air-attenuation coefficient of $3.01 \times 10^{-4} \text{ mm}^{-1}$ was used for GIWAXS. The beam centre was measured directly by applying the X-ray absorbers, and the CeO_2 sample was calibrated for the sample-to-detector distance.

Computational details

Atomistic MD simulations were performed in the GROMACS (version 2020.6) simulation package using the General Amber Force Field (gaff2) force field⁴⁵. The molecules were geometry-optimized by DFT calculations using the b3lyp/6-31g(d) method, and the RESP charge was calculated in the Multiwfn program⁴⁶. The molecules were first aligned in a 12×12 matrix and placed on top of the ITO surface modelled using the universal force field, which cover the whole periodic table. After thousands of steps of energy minimization, an equilibration of 10 ns was performed at 300 K with position restraints on the heavy atoms. The production runs extended for 100 ns under the canonic ensemble while recording every 10 ps. The temperature was coupled to 300 K using the Nose-Hoover method. A cutoff scheme of 1.2 nm was implemented for the non-bonded interactions, and the Particle Mesh Ewald method with a Fourier spacing of 0.1 nm was applied for the long-range electrostatic interactions⁴⁷. All covalent bonds with hydrogen atoms were constraint using the LINCS algorithm⁴⁸.

DFT calculations were performed with the Vienna Ab initio Simulation Package (VASP, 5.4.4 version using the projector augmented wave pseudopotential method⁴⁹, which is based on the PBE exchange-correlation functional⁵⁰). The planewave basis sets were converged at a kinetic energy cutoff of 400 eV for each slab model, and all structures were optimized until the force on each atom was less than 0.02 eV Å^{-1} . The a-SAM was optimized in a $25 \times 25 \times 25 \text{ Å}^3$ unit cell. To further investigate the interaction between defects in the perovskites and a-SAM, we constructed a $2 \times 3 \times 3$ slab model of the (001) surface of FAPbI_3 with a $(3 \times 3 \times 1)$ k -point grid.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this Article.

Data availability

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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Author contributions

X.W. and Y.H. conceived the idea and designed the experiments. Y.H. supervised the project. X.Y. synthesized and characterized the target SAM molecule. X.W. and J.L. fabricated solar cells. L.D., D.G., K.J. and S.D.S. performed hyperspectral microscopy and TRPL characterization. R.G., G.P. and P.M.-B. conducted GIWAXS characterization. X.J. conducted Hr-EQE measurements. R.L. performed the DFT calculations. R.G., H.L., X.J., J.C., Z.S., S.L., Y.W., T.W. and Q.Z. assisted with device fabrication and characterizations. X.W., R.G., J.L. and Y.H. analysed the data and wrote the paper. Y.H., X.W., R.G., J.L., P.G. and P.M.-B. reviewed and edited the paper. All authors read and commented on the paper.

Competing interests

S.D.S. is a co-founder of Swift Solar Inc. Y.H. is a founder of Singfilm Solar. The other authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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• Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	The area of tested solar cells is 1.007 cm ² .
Method used to determine the device area	☒ Yes ☐ No	Metal aperture mask (black color)

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	We provide certified J-V in both forward and backward direction in Supplementary Fig. 21.
Voltage scan conditions <i>For instance: scan direction, speed, dwell times</i>	☒ Yes ☐ No	See details in Method.
Test environment <i>For instance: characterization temperature, in air or in glove box</i>	☒ Yes ☐ No	See details in Method.
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	No preconditioning condition is used.
Stability of the J-V characteristic <i>Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.</i>	☒ Yes ☐ No	We provided MPPT data in Fig.4e

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☐ Yes ☒ No	Negligible hysteresis was observed.
Related experimental data	☒ Yes ☐ No	Reverse and forward JV scan can be found in Supplementary Fig. 21

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	We provide EQE measurement in Fig. 4f
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	We compare the integrated Jsc with one from JV scan. The difference between the integrated Jsc from EQE and Jsc from JV scan is within 2.6 % difference, which is within accuracy confidence of the measurements. Integrated Jsc from EQE is higher than Jsc from J-V scan.
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	We didn't fabricate tandem solar cell.

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	See details in Method.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Our solar simulator was calibrated by Si reference cell.(certificated by Newport.)

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes
☐ No

The champion devices were sent to do certification, the certified device spectra mismatch factor (1.004) can be found in Supplementary Fig. 21.

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

An optical aperture mask (1.007 cm²) was used.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes
☒ No

No significant variations was observed.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes
☐ No

We provide certification data in Supplementary Fig. 21.

A copy of any certificate(s)

Provide in Supplementary Information

☒ Yes
☐ No

We provide certification data in Supplementary Fig.21.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

See details in Fig. S20

Statistical analysis of the device performance

☒ Yes
☐ No

We provide statistics in Fig.4c and Supplementary Fig. 20

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature

☒ Yes
☐ No

We provided stability data in Fig.4g and h.